Lecture 8

Factoring

Objectives. *Lecture 7 read an eigenphase with a circuit. This lecture asks:* how does a phase become a factor? *Factoring reduces, by a gcd, to the order r of a random a modulo N (Proposition 8.2). Phase estimation of the multiplication by a, started in $|1\rangle$, reads a fraction s/r, and continued fractions turn it into r (Theorem 8.9). Together they find a factor of an n-bit N with circuits of $O(n^3)$ gates (Theorem 8.4), where no classical algorithm of polynomial time is known.*

By the end of this lecture you will be able to:

- Prove that $\gcd(a^{r/2}-1, N)$ is a proper factor of N when r is even and $a^{r/2} \not\equiv -1$, and state how often a random a qualifies.
- Prove that $|1\rangle$ is the uniform superposition of the eigenvectors of U_a, and deduce what phase estimation reads on it.
- Prove that $3n$ runs with $t = 2n+1$ counting qubits, continued fractions and a classical test find r with probability at least $2/3$.
- State the status of factoring: what is proved, what is cited and what is conjectured.

8.1 The reduction, its gcd, preprocessing and probability

Factoring asks, for a composite N, for a *proper factor*: a divisor d of N with $1 < d < N$. In this lecture N is the number to factor, no longer 2^n as in Lecture 7; it has n bits. Two cases are classical and fast: an even N has the factor 2, and a power $N = b^k$, $k \geq 2$, the factor b, the integer k-th root of N for some $k < n$ (Nielsen and Chuang, Ex. 5.17). A prime N, which has none, is recognised in time polynomial in n by the Miller–Rabin test (Kitaev, Shen and Vyalyi, §4.2). What remains is an odd N with at least two distinct prime factors.

Choose a uniformly in $\{1, \ldots, N-1\}$. Euclid's algorithm gives $\gcd(a, N)$ in time polynomial in n; if it is not 1, it is a proper factor. Extended, it gives integers u and v with $ua + vN = \gcd(a, N)$ (Kitaev, Shen and Vyalyi, p. 40). So an a coprime to N, a *unit* modulo N, has an *inverse* u: $ua \equiv 1$.

Definition 8.1 (The order)**.** For a coprime to N, the *order* of a modulo N is the least $r \geq 1$ with $a^r \equiv 1 \pmod{N}$.

It exists, and $r < N$: the powers $a^0, \ldots, a^{N-1}$ are units, of which there are at most $N-1$, so two agree, $a^i \equiv a^j$ with $i < j$, and multiplying by the inverse of a^i gives $a^{j-i} \equiv 1$. (This r is not the Bloch vector of *Quantum Mechanics*.)

Proposition 8.2 (The reduction · *proved modulo extended Euclid*)**.** *Let a be coprime to N, with order r. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2}-1, N)$ and $\gcd(a^{r/2}+1, N)$ are proper factors of N.*

Proof. Let $b = a^{r/2} \bmod N$. Then N divides $b^2 - 1 = (b-1)(b+1)$. It divides neither factor: $b \not\equiv 1$, as $r/2 < r$ and r is the least, and $b \not\equiv -1$ by hypothesis. If $\gcd(b-1, N)$ were 1, multiplying $(b-1)(b+1) \equiv 0$ by the inverse of $b-1$ would give $b+1 \equiv 0$; if it were N, N would divide $b-1$. So $1 < \gcd(b-1, N) < N$, and likewise for $b+1$. ∎

Theorem 8.3 (A random base · *cited*)**.** *Let N be odd with $m \geq 2$ distinct prime factors, and let a be uniform among the units modulo N. Then r is even and $a^{r/2} \not\equiv -1$ with probability at least $1 - 2^{-(m-1)} \geq \frac{1}{2}$.*

An a uniform in $\{1, \ldots, N-1\}$, once it is coprime to N, is uniform among the units. It is Kitaev, Shen and Vyalyi's (pp. 121–122) and Shor's (p. 16). Nielsen and Chuang's Theorem 5.3 (p. 233) prints $1 - 2^{-m}$, which is false: for $N = 21$ the probability is exactly $\frac{1}{2}$.

8.2 U_a, its eigenvectors, $|1\rangle$ as their superposition

The whole algorithm, Shor's, is the following. Its one quantum step, order finding, takes this section and the next two; the proof ends in §8.5.

Theorem 8.4 (Factoring · *proved modulo the cited lemmas*)**.** *For a composite n-bit N, the following finds a proper factor with probability at least $2/3$. Return 2 if N is even, and b if $N = b^k$, $k \geq 2$. Otherwise, up to three times: draw at most seven uniform n-bit strings and let a be the first with $1 \leq a < N$, skipping the try if there is none; return $\gcd(a, N)$ if it is not 1; else run Theorem 8.9; if it returns an even r, return $\gcd(a^{r/2} - 1, N)$ if that is a proper factor. It uses $9n$ circuits of $O(n^3)$ gates over the alphabet, and classical work polynomial in n.*

The draws make a uniform in $\{1, \ldots, N-1\}$ in bounded time: an odd N with n bits has $N - 1 \geq 2^{n-1}$, so a draw qualifies with probability at least $\frac{1}{2}$, and seven find a with probability at least $1 - 2^{-7}$.

For a coprime to $N < 2^n$, let U_a be the gate on n qubits with $U_a |y\rangle = |ay \bmod N\rangle$ for $y < N$, and $U_a |y\rangle = |y\rangle$ for $N \leq y < 2^n$. As a has an inverse modulo N, $y \mapsto ay \bmod N$ permutes $\{0, \ldots, N-1\}$, so U_a permutes the basis states and is unitary. On $|1\rangle$ it walks the orbit

$$|1\rangle \mapsto |a\rangle \mapsto \left|a^2\right\rangle \mapsto \cdots \mapsto \left|a^{r-1}\right\rangle \mapsto |1\rangle ,$$

with the powers taken modulo N: r distinct basis states, which U_a shifts round a cycle of length r. Its eigenvectors there are built from the characters of $\mathbb{Z}_r$, as the columns of Lecture 7's QFT were from those of $\mathbb{Z}_N$.

Lemma 8.5 (The eigenvectors of U_a)**.** *For $s = 0, \ldots, r-1$ let*

$$|u_s\rangle = r^{-1/2} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} \left|a^k\right\rangle .$$

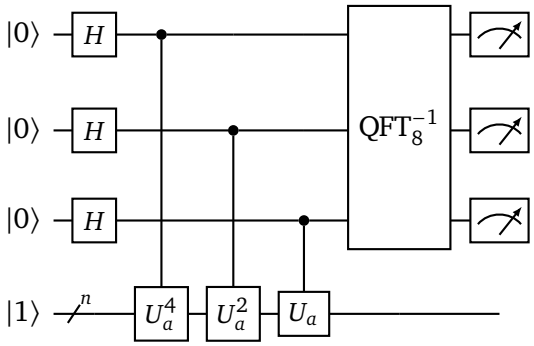
(i) $U_a |u_s\rangle = e^{2\pi i s/r} |u_s\rangle$.
(ii) The $|u_s\rangle$ are orthonormal.
(iii) $r^{-1/2} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Proof. (i) U_a maps $\left|a^k\right\rangle$ to $\left|a^{k+1}\right\rangle$, and $|a^r\rangle = \left|a^0\right\rangle$, so it shifts the index k by 1 modulo r. As $e^{-2\pi i s k/r}$ has the period r in k,

$$\begin{aligned} U_a |u_s\rangle &= r^{-1/2} \sum_k e^{-2\pi i s (k-1)/r} \left|a^k\right\rangle \\ &= e^{2\pi i s/r} |u_s\rangle . \end{aligned}$$

(ii) The $|a^k\rangle$, $k < r$, are orthonormal, so $\langle u_s|u_{s'}\rangle = r^{-1}\sum_k e^{2\pi i(s-s')k/r}$, which the geometric sum of Lecture 7 makes 1 if $s = s'$ and 0 otherwise. (iii) The sum is $r^{-1}\sum_k\big(\sum_s e^{-2\pi isk/r}\big)\,|a^k\rangle$, and the inner sum is r for $k = 0$ and 0 for $0 < k < r$. ∎

We cannot prepare $|u_s\rangle$, which needs r. But $|1\rangle$ needs only an X on the bottom wire, and by (iii) it is the uniform superposition of the $|u_s\rangle$. So run phase estimation with $U = U_a$ on $|1\rangle$; for $t = 3$,



By Theorem 7.5, whose proof uses only that the $|u_s\rangle$ are orthonormal eigenvectors, the reading y has the probability $r^{-1}\sum_s p_{s/r}(y)$, with $p_{s/r}$ the distribution of Theorem 7.3 for the phase s/r. The circuit acts as if s were drawn uniformly from $\{0,\dots,r-1\}$ and s/r then estimated. The target is never read.

Two guards of earlier lectures bite on the controlled powers. They must control U_a itself, exactly, not a gate equal to it up to a global phase (Example 6.7). And a workspace they use must end in $|0\cdots0\rangle$: left depending on y, it would tell the branches apart, as garbage did (Example 5.2).

8.3 Precision $t = 2n + 1$ and continued fractions

Fix s. Call the reading *good* if it is within 2^{-t-1} of s/r in the circular distance. By Theorem 7.3 it is good with probability at least $4/\pi^2$, and at least $4/\pi^2 - \delta$ when $\mathrm{QFT}_{2^t}^{-1}$ is compiled *within* δ, moving no probability by more than δ. Take $t = 2n + 1$. On a good reading the distance is the plain one: for $s = 0$ the phase 0 is read exactly, as $y = 0$; for $s \neq 0$, s/r is at least $1/r > 2^{-t-1}$ from 0 and from 1, so no good reading wraps round. And as $r < N < 2^n$,

$$\left|\frac{y}{2^t} - \frac{s}{r}\right| \le 2^{-2n-2} < \frac{1}{2r^2}.$$

A rational x has a continued fraction

$$x = c_0 + \cfrac{1}{c_1 + \cfrac{1}{\ddots + \cfrac{1}{c_m}}},$$

with integers $c_0 = \lfloor x \rfloor$ and $c_1,\dots,c_m \ge 1$, found by Euclid's algorithm on its numerator and denominator. Its *convergents* are the fractions obtained by stopping at $c_0, c_1, \dots, c_m$ in turn. For $x = y/2^t$ there are $O(t)$ of them, as Euclid's algorithm at least halves the remainder every two steps.

Theorem 8.6 (Continued fractions · *cited*). *Let x be rational and p/q a fraction in lowest terms, $q \ge 1$, with $|x - p/q| < 1/(2q^2)$. Then p/q is a convergent of x.*

This is Nielsen and Chuang's Theorem 5.1 (p. 229, proof pp. 637–638), and Kitaev, Shen and Vyalyi's Theorem A.13 (p. 250). Nielsen and Chuang print $\leq$; for the expansion Euclid's algorithm gives, that fails at $x = \frac{1}{2}$ and $p/q = \frac{1}{1}$, as the convergents of $\frac{1}{2}$ are $\frac{0}{1}$ and $\frac{1}{2}$.

On a good reading, write s/r in lowest terms as s'/r', with r' a divisor of r. Then $|y/2^t - s'/r'| < 1/(2r^2) \leq 1/(2r'^2)$, so s'/r' is a convergent of $y/2^t$, with $r' \leq r < N$. A run therefore tests the denominators $q < N$ of the convergents of $y/2^t$; r' is among them, and $r' = r$ exactly when $\gcd(s, r) = 1$.

For example, the order of 2 modulo 21 is 6: the orbit is 1, 2, 4, 8, 16, 11. Here $n = 5$ and $t = 11$, and for $s = 1$ the good reading is $y = 341$, as $2^{11}/6 \approx 341.3$. Euclid's algorithm gives $341/2048 = [0; 6, 170, 2]$, with the convergents $\frac{0}{1}$, $\frac{1}{6}$, $\frac{170}{1021}$ and $\frac{341}{2048}$. The denominators below 21 are 1 and 6, and $2^6 = 64 \equiv 1 \pmod{21}$, while $2^1 \not\equiv 1$. Then $2^3 = 8 \not\equiv -1$, and $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$.

8.4 Coprimality, verification, repetitions

A run finds r itself when its reading is good and $\gcd(s, r) = 1$. How likely is the second, and how is r recognised?

Proposition 8.7 (Coprimality · *proved modulo Euler's product*)**.** *Let $r \geq 1$ have k distinct prime factors, and let s be uniform in $\{0, \dots, r-1\}$. Then*

$$\begin{aligned} \Pr\big[\gcd(s, r) = 1\big] &= \left|\mathbb{Z}_r^\times\right| / r \\ &= \textstyle\prod_{p \mid r} (1 - 1/p) \\ &\geq 1/(k+1). \end{aligned}$$

If $r < 2^n$, the probability is at least $1/n$.

Proof. $\left|\mathbb{Z}_r^\times\right|$ counts the $s < r$ coprime to r, and the product is Euler's (Nielsen and Chuang, (A4.23), p. 631). List the primes dividing r as $p_1 < \cdots < p_k$. The i-th prime number is at least $i+1$, so $p_i \geq i+1$ and the product is at least $\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{k}{k+1} = \frac{1}{k+1}$. Finally $r \geq p_1 \cdots p_k \geq 2^k$, so $r < 2^n$ gives $k + 1 \leq n$. ∎

Lemma 8.8 (Verification)**.** *If $a^d \equiv 1 \pmod{N}$ with $d \geq 1$, then r divides d.*

Proof. Write $d = qr + \rho$ with $0 \leq \rho < r$. Then $1 \equiv a^d = (a^r)^q a^\rho \equiv a^\rho$, and $\rho \geq 1$ would contradict the minimality of r. ∎

So the test $a^q \equiv 1 \pmod{N}$, by repeated squaring, passes r and its multiples, and none of its proper divisors. In the example, $s = 2$ gives $\frac{1}{3}$, and $2^3 \not\equiv 1 \pmod{21}$. A bad reading can pass a multiple of r: a passing q shows only that r divides it, so we keep the least.

Theorem 8.9 (Order finding · *proved modulo continued fractions and Euler's product*)**.** *Let a be coprime to $N < 2^n$, with order r. Run phase estimation of U_a on $|1\rangle$ with $t = 2n + 1$, its $\mathrm{QFT}_{2^t}^{-1}$ compiled within $\delta = 1/200$, $3n$ times; for each reading y test every denominator $q < N$ of a convergent of $y/2^t$, and return the least q that passes, or nothing if none does. The return is r with probability at least $2/3$.*

Proof. For each s the reading is good with probability at least $4/\pi^2 - \delta \geq 2/5$, and s is uniform, so by Proposition 8.7 a run passes r with probability at least $2/(5n)$. By Lemma 8.8

every passing q is a multiple of r, so the least is r once some run passes r. All $3n$ runs miss with probability at most $(1 - 2/(5n))^{3n} \le e^{-6/5} < 1/3$. ∎

8.5 Modular powers

Phase estimation needs $\mathrm{C}(U_a^{2^j})$ for $j < t$. From $\mathrm{C}(U_a)$ that is $2^t - 1$ uses (Lecture 7). But $U_a^{2^j} = U_{b_j}$ with $b_j = a^{2^j} \bmod N$, computed classically by j squarings: one controlled multiplication by a known constant each.

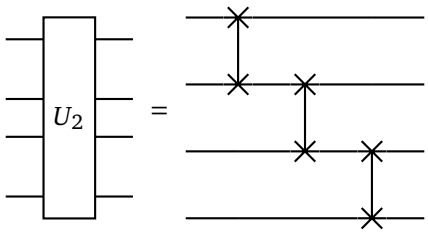
Theorem 8.10 (Modular powers · *cited*)**.** *For b coprime to $N < 2^n$, $\mathrm{C}(U_b)$ has an exact circuit of $O(n^2)$ gates over the alphabet, with a workspace of $O(n)$ qubits that starts and ends in $|0\cdots0\rangle$.*

The circuits are reversible as in Lecture 1 and exact as the Toffoli is ((6.2)); see Nielsen and Chuang, Box 5.2 and p. 232, Shor, pp. 10–11, and Watrous, p. 221.

Proof of Theorem 8.4. What reaches the loop has $m \ge 2$ distinct primes, and a try finds a with probability at least $1 - 2^{-7}$. It then succeeds when Theorem 8.3 and Theorem 8.9 both do, with probability at least $\frac{1}{2} \cdot \frac{2}{3}$, by Proposition 8.2; three tries fail with probability at most $(1 - \frac{127}{384})^3 < 1/3$. A try runs $3n$ circuits of $O(n^3)$ gates: t controlled powers of $O(n^2)$ gates each, t Hadamards, and $\mathrm{QFT}_{2^t}^{-1}$ compiled within $1/200$ in $O(n^2 \log^c n)$ gates (Theorem 7.2). ∎

8.6 F15, in short

For $N = 15$ and $a = 2$ the orbit of 1 is 1, 2, 4, 8, so $r = 4$, and $2^2 = 4 \not\equiv -1$: $\gcd(3, 15) = 3$ and $\gcd(5, 15) = 5$. As $16 \equiv 1 \pmod{15}$, doubling a 4-bit y modulo 15 moves its top bit to the bottom, $y_1y_2y_3y_4 \mapsto y_2y_3y_4y_1$ ($y = 15$ is fixed either way):



(8.1)

an equality of matrices. So $U_4 = U_2^2$ is two swaps, $U_2^4 = \mathbb{1}$, and each controlled power is a few controlled swaps, exact. The phases $s/4$ are read exactly for every $t \ge 2$: $s = 1$ and $s = 3$ give $r = 4$, $s = 2$ only the divisor 2, which fails the test, and $s = 0$ nothing. An instance illustrates and proves nothing about scaling; $r = 4$ divides 2^t, and the order of 2 modulo 21 above, like Lecture 7's phase $\frac{1}{3}$, shows the general case.

8.7 Status

Theorem 8.4 is about a function problem, and states an algorithm with its resources: bounded error, $O(n)$ circuits of $O(n^3)$ gates. It is proved modulo the cited lemmas. No classical algorithm of polynomial time is known; the best known, the number field sieve, runs in time $\exp(O(n^{1/3}(\log n)^{2/3}))$ by a heuristic analysis (Nielsen and Chuang, p. 216;

de Wolf, p. 37). That none exists is a conjecture, on which much public-key cryptography rests. This is the evidence Lecture 6 promised for its conjecture BQP ≠ BPP: a quantum algorithm proved, a classical one not known. Unlike Simon's separation (Lecture 5), it needs no oracle, but no classical lower bound is proved.

Summary.

- *The reduction:* if r is even and $a^{r/2} \not\equiv -1$, $\gcd(a^{r/2} \pm 1, N)$ are proper factors (Proposition 8.2); a random a qualifies with probability at least $\frac{1}{2}$, cited.
- *The phase:* $|1\rangle$ is the uniform superposition of the eigenvectors of U_a, with the phases s/r (Lemma 8.5).
- *The main result:* $t = 2n + 1$, continued fractions, the coprimality bound and the least passing denominator give r with probability at least $2/3$ in $3n$ runs (Theorem 8.9), and a factor (Theorem 8.4).
- *Status:* proved modulo cited lemmas; classical hardness is conjectured.
- *Next:* Lecture 9, search.

References

Nielsen and Chuang, §5.3, pp. 226–235, Box 5.2, p. 228, Theorem A4.16, pp. 637–638, and p. 216. Kitaev, Shen and Vyalyi, §4.2, pp. 38–41, §13.3, pp. 120–122, and Theorem A.13, p. 250. Watrous (2025), pp. 218–228. De Wolf, pp. 37–38. P. W. Shor, *Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer*, arXiv:quant-ph/9508027 (1995), pp. 8–16.